

Algebra II Counting, Probability, and Statistics Cheatsheet

Module 13

Counting Principle

If one choice has m options and the next has n options, then together there are

$$mn$$

outcomes.

Factorials

$$n! = n(n-1)(n-2) \cdots 1$$

$$0! = 1$$

Permutations and Combinations

Order matters:

$${}_nP_r = \frac{n!}{(n-r)!}$$

Order does not matter:

$${}_nC_r = \frac{n!}{r!(n-r)!}$$

Repeated arrangements:

$$\frac{n!}{a!b!c!\cdots}$$

Probability Rules

$$P(A) = \frac{\text{favorable outcomes}}{\text{total outcomes}}$$

Complement:

$$P(A^c) = 1 - P(A)$$

Addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Odds:

$$\text{odds in favor} = \frac{\text{favorable outcomes}}{\text{unfavorable outcomes}}$$

Independence

Events A and B are independent when:

$$P(A \mid B) = P(A)$$

Equivalent test:

$$P(A \cap B) = P(A)P(B)$$

Statistics

$$\text{mean} = \frac{\sum x}{n}$$

$$\text{range} = \text{max} - \text{min}, \quad \text{IQR} = Q_3 - Q_1$$

Z-score:

$$z = \frac{x - \mu}{\sigma}$$

Data value from z-score:

$$x = \mu + z\sigma$$

Normal Model Reminders

Empirical rule:

$$68\% \text{ within } 1\sigma, \quad 95\% \text{ within } 2\sigma, \quad 99.7\% \text{ within } 3\sigma$$

Quick Checks

- Use permutations when positions or order matter.
- Use combinations when only the group matters.
- Conditional probability changes the denominator.
- Independence is a probability relationship, not a wording clue.